MATH744 - Lie Groups I



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1 Matrix groups

Definition 1.1 The exact sequence

$$1 \longrightarrow A \longrightarrow B \underset{s}{\overset{\phi}{\rightleftarrows}} C \longrightarrow 1$$

split if there exists $s : C \to B$ such that $\phi s = 1_C$ which is equivalent of saying that $B = C \ltimes A$

Lemma 1.2

- $e^0 = I$
- $(e^X)^T = e^{X^T}$
- $\overline{e^X} - e^{\overline{X}}$
- $(e^X)^{-1} = e^{-X}$
- $ge^X g^{-1} = e^{gXg^{-1}}$
- $\frac{d}{dt}e^{tX} = Xe^{tX}$

Theorem 1.3 $\phi_X : \mathbb{R} \to \mathrm{GL}(n, \mathbb{C})$, $t \mapsto e^{tX}$ is a group homomorphism. Conversely, any group homomorphism $\phi : \mathbb{R} \to \mathrm{GL}(n, \mathbb{C})$ is of the form ϕ_X for some $X \in M_n(\mathbb{C})$

Definition 1.4 G is a Lie group, the Lie algebra is

$$\begin{aligned}\mathrm{Lie}(G) &= \{X \in M_n(\mathbb{C}) | e^{tX} \in G, \forall t \in \mathbb{R}\} \\ &= \{X \in M_n(\mathbb{C}) | \text{im } \phi_X \subseteq G\}\end{aligned}$$

2 Homeworks

2.1 Homework1

Exercise 2.1 Let $G = O(2,\mathbb{R})$ be the (real) orthogonal group in dimension 2

1. Show that $\mathrm{SO}(2,\mathbb{R})$ is isomorphic to the circle S^1
2. Show that $O(2,\mathbb{R})$ has two connected components, and is not abelian
3. Show that $\{g \in O(2,\mathbb{R}) | \det g = -1\}$ is a single conjugacy class
4. Show there is an exact sequence $1 \to \mathrm{SO}(2) \to O(2) \to \mathbb{Z}/2\mathbb{Z} \to 1$. Is this sequence split

Solution.

1. Suppose $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SO(2,\mathbb{R})$, then

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} a^2+b^2 & ac+bd \\ ac+bd & c^2+d^2 \end{pmatrix} \Rightarrow \begin{cases} a^2+b^2=1 \\ ac+bd=0 \\ c^2+d^2=1 \end{cases}$$

   we can find $0 \le \theta - \varphi \le 2\pi$ such that $\begin{cases} a=\cos\theta \\ b=\sin\theta \\ c=\cos\varphi \\ d=\sin\varphi \end{cases}$

   then $0 = \cos\theta\cos\varphi + \sin\theta\sin\varphi = \cos\theta - \varphi$, hence $\theta - \varphi = \frac{\pi}{2}$ or $\frac{3\pi}{2}$

   hence if $\theta - \varphi = \frac{\pi}{2}$, $g = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$, $\det g = 1$, if $\theta - \varphi = \frac{3\pi}{2}$, $g = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$, $\det g = 1$, thus $\phi : S^1 \to SO(2,\mathbb{R})$, $\theta \mapsto \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$ is an isomorphism

2. From the analysis in (a), all matrices of form $\left\{\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}\right\} = O(2,\mathbb{R}) \setminus SO(2,\mathbb{R})$ is path-connected, and all matrices of form $\left\{\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}\right\} = SO(2,\mathbb{R})$ is also path-connected, but $O(2,\mathbb{R})$ is not path-connected since $\det : O(2,\mathbb{R}) \to \mathbb{R}$ is a continuous function but the image is $\{\pm 1\}$ is not connected, thus $O(2,\mathbb{R})$ has two connected components

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} \neq \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

   hence $O(2,\mathbb{R})$ is not abelian

3. $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$ hence $O(2,\mathbb{R}) \setminus SO(2,\mathbb{R})$ is a single conjugacy class
4. Notice $\det : O(2,\mathbb{R}) \to \mathbb{Z}/2\mathbb{Z} \cong \{\pm 1, \text{multiplication}\}$ is a surjective group homomorphism with kernel $SO(2,\mathbb{R})$, hence

$$1 \longrightarrow SO(2,\mathbb{R}) \hookrightarrow O(2,\mathbb{R}) \xrightarrow{\det} \mathbb{Z}/2\mathbb{Z} \longrightarrow 1$$

   is an exact sequence

   There is a map $\mu : \mathbb{Z}/2\mathbb{Z} \to O(2,\mathbb{R})$, $\bar{1} \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ which is a homomorphism such that $\det \circ \mu = 1$, hence the exact sequence splits

□

Exercise 2.2

(a)

Suppose $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SO(1,1)$, then

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} a^2+b^2 & ac+bd \\ ac+bd & c^2+d^2 \end{pmatrix} \Rightarrow \begin{cases} a^2-b^2=1 \\ ac-bd=0 \\ c^2-d^2=-1 \end{cases}$$

we can find $x, y \in \mathbb{R}$ such that $\begin{cases} b = \sinh x \\ c = \sinh y \end{cases}$, and $\begin{cases} a = \cosh x \\ d = \cosh y \end{cases}$ or $\begin{cases} a = -\cosh x \\ d = \cosh y \end{cases}$ or $\begin{cases} a = \cosh x \\ d = -\cosh y \end{cases}$ or $\begin{cases} a = -\cosh x \\ d = -\cosh y \end{cases}$

then

$$0 = \cosh x \sinh y - \sinh x \cosh y = \sinh(y - x) \Leftrightarrow y = x, \text{or}$$

$$0 = -\cosh x \sinh y - \sinh x \cosh y = -\sinh(x + y) \Leftrightarrow y = -x, \text{or}$$

$$0 = \cosh x \sinh y + \sinh x \cosh y = \sinh(x + y) \Leftrightarrow y = -x, \text{or}$$

$$0 = -\cosh x \sinh y + \sinh x \cosh y = \sinh(x - y) \Leftrightarrow y = x$$

then $\det g = 1, -1, -1, 1$ correspondingly, hence g can only be $\begin{pmatrix} \cosh x & \sinh x \\ \sinh x & \cosh x \end{pmatrix}$ or $\begin{pmatrix} -\cosh x & \sinh x \\ \sinh x & -\cosh x \end{pmatrix}$, consider $\phi : \mathbb{R}^* \to SO(1,1)$,

$$x \mapsto \begin{cases} \begin{pmatrix} \cosh(\ln x) & \sinh(\ln x) \\ \sinh(\ln x) & \cosh(\ln x) \end{pmatrix}, x > 0 \\ \begin{pmatrix} -\cosh(\ln(\frac{1}{-x})) & \sinh(\ln(\frac{1}{-x})) \\ \sinh(\ln(\frac{1}{-x})) & -\cosh(\ln(\frac{1}{-x})) \end{pmatrix}, x < 0 \end{cases}$$

is an isomorphism easily by checking 4 cases

(b)

Notice $\det : O(1,1) \to \mathbb{Z}/2\mathbb{Z} \cong \{\pm 1, \text{multiplication}\}$ is a surjective group homomorphism with kernel $SO(1,1)$, hence $O(1,1)/SO(1,1) \cong \mathbb{Z}/2\mathbb{Z}$

(c)

Notice ϕ from (a) is continuous, hence $O(1,1)^0 = \left\{ \begin{pmatrix} \cosh x & \sinh x \\ \sinh x & \cosh x \end{pmatrix} \right\}$, and $\frac{x}{|x|} : \mathbb{R}^* \to \mathbb{Z}/2\mathbb{Z} \cong \{\pm 1, \text{multiplication}\}$ is a surjective group homomorphism with kernel $\mathbb{R}_{>0}$, hence $SO(1,1)/O(1,1)^0 \cong \mathbb{R}^*/\mathbb{R}_{>0}$, $\left|O(1,1)/O(1,1)^0\right| = [O(1,1) : SO(1,1)][SO(1,1) : O(1,1)^0] = 4$, hence it is isomorphic to either $\mathbb{Z}/4\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, but $-I, J, -J \in O(1,1)/O(1,1)^0$ are of order 2 and $(-I)J = -J, (-I)(-J) = J, J(-J) = -I$, thus the component group $O(1,1)/O(1,1)^0 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$

Exercise 2.3

(a)

Here $J \in GL(n, \mathbb{C})$, $G_J = \{g \in GL(n, \mathbb{C}) | g^T J g = J\} = \{g \in GL(n, \mathbb{C}) | g^T = J g^{-1} J^{-1}\}$, $G_I = \{g \in GL(n, \mathbb{C}) | g^T g = I\} = \{g \in GL(n, \mathbb{C}) | g^T = g^{-1}\}$, since J is symmetric, $\exists P \in GL(n, \mathbb{C})$ such that $PJP^T = I$(corresponding to changing of basis), then we have $PG_J P^{-1} = G_I$ since $\forall g \in G_J$, we have $Jg^T J^{-1} = g^{-1}$, then $(PgP^{-1})^{-1} = Pg^{-1}P^{-1} = PJg^T J^{-1} P^{-1} = PJP^T (P^T)^{-1} g^T P^T (P^T)^{-1} J^{-1} P^{-1} = (PJP^T)(PgP^{-1})^T (PJP^T)^{-1} = (PgP^{-1})^T$, hence $PgP^{-1} \in G_I$, thus G_J and $O(n, \mathbb{C})$ are conjugate in $GL(n, \mathbb{C})$

(b)

Let $K = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$, then $\begin{pmatrix} a & \\ & b \end{pmatrix} \begin{pmatrix} & 1 \\ 1 & \end{pmatrix} \begin{pmatrix} a & \\ & b \end{pmatrix} = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$ which gives $ab = 1$, if n is even, let $J = \begin{pmatrix} K & & \\ & \ddots & \\ & & K \end{pmatrix}$, then the diagonal subgroup of G_J is isomorphic to $\mathbb{C}^{*\frac{n}{2}} = \mathbb{C}^{*\lfloor \frac{n}{2} \rfloor}$, if n is odd, if $J = \begin{pmatrix} K & & & \\ & \ddots & & \\ & & K & \\ & & & 0 \end{pmatrix}$, then the diagonal subgroup of G_J is isomorphic to $\mathbb{C}^{*\lfloor \frac{n}{2} \rfloor}$, if $J = \begin{pmatrix} K & & \\ & \ddots & \\ & & K \\ & & & 1 \end{pmatrix}$, then the diagonal subgroup of G_J is isomorphic to $\mathbb{C}^{*\lfloor \frac{n}{2} \rfloor} \times \mathbb{Z}/2\mathbb{Z}$ which contains a subgroup isomorphic to $\mathbb{C}^{*\lfloor \frac{n}{2} \rfloor}$

(c)

By (a) and (b), we know that $O(n, \mathbb{C})$ contains a subgroup isomorphic to $\mathbb{C}^{*\lfloor \frac{n}{2} \rfloor}$

(d)

If $A \in \text{Lie}$, then $e^{tA} \in O(n,\mathbb{C})$, $e^{tA^T} = \left(e^{tA}\right)^T = \left(e^{tA}\right)^{-1} = e^{-tA}$, then $A^T = \frac{de^{tA^T}}{dt}|_{t=0} = \frac{de^{-tA}}{dt}|_{t=0} = -A$, hence $\text{Lie}(O(n,\mathbb{C})) = \left\{A \in M_n(\mathbb{C}) \parallel A^T = -A\right\}$, $\dim \text{Lie}(O(n,\mathbb{C})) = \frac{n(n-1)}{2}$

Exercise 2.4

(a)

$$1 \longrightarrow SL(n,\mathbb{C}) \hookrightarrow GL(n,\mathbb{C}) \xrightarrow{\det} \mathbb{C}^* \longrightarrow 1$$

is an exact sequence

(b)

$$1 \longrightarrow \mathbb{C}^* \xhookrightarrow{a \mapsto aI} GL(n,\mathbb{C}) \xrightarrow{q} PGL(n,\mathbb{C}) \longrightarrow 1$$

is an exact sequence, where q is the quotient map

(c)

$SL(n,\mathbb{C}) \to GL(n,\mathbb{C})$ pass to $PSL(n,\mathbb{C}) \to PGL(n,\mathbb{C})$ by quotient map, for $AZ \in PGL(n,\mathbb{C})$, $\frac{A}{\sqrt[n]{\det A}} Z \mapsto AZ$, hence the map is surjective, if $AZ \in PSL(n,\mathbb{C})$, and $A \in Z(PGL(n,\mathbb{C}))$, then $A = aI$, for some $a \neq 0$, but since $A \in SL(n,\mathbb{C})$, $1 = \det A$, $A \in Z(PSL(n,\mathbb{C}))$, hence the map is also injective

(d)

When n is odd, the previous argument still works, namely $PSL(n,\mathbb{R}) \cong PGL(n,\mathbb{R})$, when $n = 2k$ is even, the injectivity still works, however, for $aI, A \in GL(n,\mathbb{R})$, $\det(aI \cdot A) = a^{2k} \det A$ has the same sign as $\det A$, thus there is an exact sequence

$$1 \longrightarrow PSL(n,\mathbb{R}) \hookrightarrow PGL(n,\mathbb{R}) \xrightarrow{\det} \mathbb{Z}/2\mathbb{Z} \longrightarrow 1$$

Exercise 2.5

(a)

If $g = \begin{pmatrix} \alpha & \beta \\ \gamma & \xi \end{pmatrix} \in SU(2)$, then

$$I = gg^* = \begin{pmatrix} \alpha & \beta \\ \gamma & \xi \end{pmatrix} \begin{pmatrix} \bar{\alpha} & \bar{\gamma} \\ \bar{\beta} & \bar{\xi} \end{pmatrix} = \begin{pmatrix} |\alpha|^2 + |\beta|^2 & \alpha\bar{\gamma} + \beta\bar{\xi} \\ \bar{\alpha}\gamma + \bar{\beta}\xi & |\gamma|^2 + |\xi|^2 \end{pmatrix}$$

Hence $\begin{cases} |\alpha|^2 + |\beta|^2 = 1 \\ \alpha\bar{\gamma} + \beta\bar{\xi} = 0 \\ |\gamma|^2 + |\xi|^2 \end{cases}$, and also $1 = \det g = \alpha\xi - \beta\gamma$

If $\gamma = 0$, then $|\alpha| = |\xi| = 1$, $\beta = 0$, $\alpha\xi = 1 \Rightarrow \xi = \bar{\alpha}$

If $\gamma = 0$, then $\alpha = -\frac{\beta\bar{\xi}}{\bar{\gamma}}$, $\beta = 0$, then $1 = -\frac{\beta|\xi|^2}{\bar{\gamma}} - \beta\gamma \Rightarrow -\bar{\gamma} = \beta$, hence $\alpha = \bar{\xi}$

(b)

Since $S^3 \subseteq \mathbb{C}^2 = \left\{|\alpha|^2 + |\beta|^2 = 1\right\}$ which is the same zero set, thus $SU(2)$ is topologically equivalent to S^3, and is therefore 3-dimensional, connected and simply connected

(c)

$$1 \longrightarrow SU(2) \hookrightarrow U(2) \xrightarrow{\det} S^1 \longrightarrow 1$$

is an exact sequence, there is a map $\mu : S^1 \to U(2)$, $z \mapsto \begin{pmatrix} 1 & 0 \\ 0 & z \end{pmatrix}$ which is a homomorphism such that $\det \circ \mu = 1$, hence the exact sequence splits

(d)

$$\begin{pmatrix}\alpha & \beta\\ \gamma & \xi\end{pmatrix}+\begin{pmatrix}\bar{\alpha} & \bar{\gamma}\\ \bar{\beta} & \bar{\xi}\end{pmatrix}=\begin{pmatrix}\alpha+\bar{\alpha} & \beta+\bar{\gamma}\\ \gamma+\bar{\beta} & \xi+\bar{\xi}\end{pmatrix}=\begin{pmatrix}0 & 0\\ 0 & 0\end{pmatrix}\Rightarrow \alpha+\bar{\alpha}=\xi+\bar{\xi}=\gamma+\bar{\beta}=0$$

and $0=trg=\alpha+\xi$, thus $\xi=-\alpha\in i\mathbb{R}$ and $\gamma=-\bar{\beta}$, hence W is a 3-dimensional real vector space of the form $\left\{\begin{pmatrix}ix & y+iz\\ -y+iz & -ix\end{pmatrix}\mid x,y,z\in\mathbb{R}\right\}$ and $tr\left(\begin{pmatrix}ix_1 & y_1+iz_1\\ -y_1+iz_1 & -ix_1\end{pmatrix}\begin{pmatrix}ix_2 & y_2+iz_2\\ -y_2+iz_2 & -ix_2\end{pmatrix}\right)=-2(x_1x_2+y_1y_2+z_1z_2)$ is a definite symmetric bilinear form

(e)

$\forall g\in SU(2),\quad g^*=g^{-1},\quad X,Y\in W,\quad gXg^{-1}+(gXg^{-1})^*=gXg^{-1}+gX*g^*=g(X+X^*)g^*=0,\quad tr(gXg^{-1})=tr(X)=0,\quad (gXg^{-1},gYg^{-1})=tr(gXg^{-1}gYg^{-1})=tr(gXYg^{-1})=tr(XY)=(X,Y)$, hence $SU(2)$ acts on W by conjugation, preserving the form

(f)

By calculation in (d), $\langle X,Y\rangle\coloneqq\frac{1}{2}tr(XY)$ will be the standard inner product in $\mathbb{R}^3$, and $SU(2)$ still acts on W by conjugation, preserving the form, this can be regarded as a continuous homomorphism $\varphi:SU(2)\to O(3)$, since $SU(2)$ is connected by (b), hence φ is actually a continuous homomorphism from $SU(2)$ to $SO(3)$, now compute its kernel

$$\begin{pmatrix}ix & 0\\ 0 & -ix\end{pmatrix}=\begin{pmatrix}\alpha & \beta\\ -\bar{\beta} & \bar{\alpha}\end{pmatrix}\begin{pmatrix}ix & 0\\ 0 & -ix\end{pmatrix}\begin{pmatrix}\bar{\alpha} & -\beta\\ \bar{\beta} & \alpha\end{pmatrix}=ix\begin{pmatrix}|\alpha|^2-|\beta|^2 & -2\alpha\beta\\ -2\bar{\alpha}\bar{\beta} & |\beta|^2-|\alpha|^2\end{pmatrix}\Rightarrow\begin{cases}\beta=0\\ |\alpha|^2=1\end{cases}$$

and

$$\begin{pmatrix}\ & \gamma\\ -\bar{\gamma} & \ \end{pmatrix}=\begin{pmatrix}\alpha & \ \\ \ & \bar{\alpha}\end{pmatrix}\begin{pmatrix}\ & \gamma\\ -\bar{\gamma} & \ \end{pmatrix}\begin{pmatrix}\bar{\alpha} & \ \\ \ & \alpha\end{pmatrix}=\begin{pmatrix}\ & \alpha^2\gamma\\ -\bar{\alpha}^2\bar{\gamma} & \ \end{pmatrix}\Rightarrow\alpha^2=1\Rightarrow\alpha=\pm1$$

Thus we have the exact sequence

$$1\longrightarrow\mathbb{Z}/2\mathbb{Z}\longrightarrow SU(2)\xrightarrow{\varphi}SO(3)\longrightarrow 1$$

Exercise 2.6 Let $G=\mathrm{GL}(2,\mathbb{F}_q)$

1. Show that G acts transitively on $X=\mathbb{F}_2^2-\{(0,0)\}$
2. Show that the stabilizer of a point has order $q(q-1)$
3. Show there is a bijection $G/H\leftrightarrow X$
4. Show that $|\mathrm{GL}(2,\mathbb{F}_q)|=(q+1)q(q-1)^2$
5. What is the order of $\mathrm{PGL}(2,\mathbb{F}_q)|$? How about $\mathrm{PGL}(2,\mathbb{F}_5)$
6. *Extra credit*: Show that $\mathrm{PGL}(2,\mathbb{F}_q)\cong A_5$

Solution.

1. $\forall(x,y)\in X$, if $x=0$, then $\begin{pmatrix}0 & 1\\ y & 0\end{pmatrix}\begin{pmatrix}1\\ 0\end{pmatrix}=\begin{pmatrix}0\\ y\end{pmatrix}$, if $x\neq0$, then $\begin{pmatrix}x & 0\\ y & 1\end{pmatrix}\begin{pmatrix}1\\ 0\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}$, hence $\begin{pmatrix}1\\ 0\end{pmatrix}$ can move to any point in X by an action of G, hence G acts transitively on X
2. If $\begin{pmatrix}1\\ 0\end{pmatrix}=\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}1\\ 0\end{pmatrix}=\begin{pmatrix}a\\ c\end{pmatrix}\Rightarrow\begin{cases}a=1\\ c=0\end{cases}$, and since $0\neq\det g=ad-bc=d$, hence $|H|=q(q-1)$
3. Consider $\varphi:G/H\to X$, $gH\mapsto g\begin{pmatrix}1\\ 0\end{pmatrix}$ which is well-defined, and surjectivity follows from (a), injectivity follows from the fact if $g\begin{pmatrix}1\\ 0\end{pmatrix}=g'\begin{pmatrix}1\\ 0\end{pmatrix}$, then $g^{-1}g'\begin{pmatrix}1\\ 0\end{pmatrix}=\begin{pmatrix}1\\ 0\end{pmatrix}\Rightarrow g^{-1}g'\in H$, hence φ is bijective
4. Because of (c), $|G|=|G/H||H|=|X||H|=(q^2-1)q(q-1)=(q+1)q(q-1)^2$
5. Since $Z=\{aI|a\neq0\}$, $|Z|=q-1$, $|PGL(2,\mathbb{F}_q)|=|G|/|Z|=(q+1)q(q-1)$, in particular, $|PGL(2,\mathbb{F}_5)|=120$

6. From group action $GL(2,\mathbb{F}_4)$ on $\mathbb{F}_4^2 \setminus \{(0,0)\}$, we can pass to a group action $PGL(2,\mathbb{F}_4)$ on $P\mathbb{F}_4$, this action is faithful since for $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, if $c \neq 0$, $g\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} \neq \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ in $P\mathbb{F}_4$, if $b \neq 0$, $g\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} b \\ d \end{pmatrix} \neq \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ in $P\mathbb{F}_4$, hence $b = c = 0$, if $a \neq d$, $g\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} a \\ d \end{pmatrix} \neq \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ in $P\mathbb{F}_4$, hence $g \in Z$, thus we get a injective group

$$PGL(2,\mathbb{F}_4) \hookrightarrow S_5$$

since $|P\mathbb{F}_4| = 5$, but $|PGL(2,\mathbb{F}_4)| = 60$, $|S_5| = 120$, any subgroup of S_5 of order 60 should be normal, intersection of normal subgroups is again a normal subgroup, but A_5 is a simple group, hence A_5 is the only subgroup of S_5 of order 60, thus $PGL(2,\mathbb{F}_4) \cong A_5$

Note that $G = \mathrm{PGL}(2,\mathbb{F}_5)$ has exactly 5 5-Sylow subgroups, and it acts by conjugation on this set of 5 elements. The resulting map $G \to S_5$ is injective, and has image A_5. Also $\mathrm{PGL}(2,\mathbb{F}_5)$ as on the projective plane over $\mathbb{F}_5$, which has 6 elements. This gives a homomorphism $G \to S_6$. Now S_6 has an outer automorphism (unique among symmetric groups), which takes an "obvious" S_5 subgroup (fixing one point of the 6 element set) to a subgroup not fixing any elements of the set. The even permutations in the image give A_5

□

Exercise 2.7

Solution.

1. By Jordan normal form theorem, $\forall g \in GL(n,\mathbb{C})$, g can be decomposed uniquely as $SU = US$ where S is semisimple and U is unipotent, thus $\log U = \sum_{m=1}^{\infty} \frac{(-1)^{m+1}(U-I)^m}{m}$ is actually a finite sum, so is $e^{\log U}$, now consider $U(t) = I + t(U - I)$, when t is small, $e^{\log U(t)} = U(t)$, and since $e^{\log U(t)} - U(t)$ is a matrix with entries polynomials in t, $e^{\log U(t)} = U(t), \forall t$, in particular, $e^{\log U} = U$, the general case follows by dividing g into Jordan blocks

2. The image of then exponential map from $M_n(\mathbb{R})$ to $GL(n,\mathbb{R})$ is $\{A \in GL(n,\mathbb{R}) | \det A | U > 0, U \text{ is any A invariant space}\}$, notice that $\det e^{A|U} = e^{trA|U} > 0$, $R_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = \exp\begin{pmatrix} & \theta \\ -\theta & \end{pmatrix}$, for any $A \in GL(n,\mathbb{R})$ with $\det A > 0$, A can be written in real Jordan canonical form, and then a Jordan block can be written as $SU = US$, U is unipotent, and S is diagonalizable with real numbers or R_θ's, for negative diagonal numbers we can pair them up and notice $R_\pi = \begin{pmatrix} -1 & \\ & -1 \end{pmatrix} = \exp\begin{pmatrix} & \pi \\ -\pi & \end{pmatrix}$, thus A is in the image of the exponential map

□

2.2 Homework2

Exercise 2.8 Show that if $\phi : \mathfrak{g} \to \mathfrak{g}'$ is a Lie algebra homomorphism then $\ker(\phi)$ is an ideal in $\mathfrak{g}$. How about the converse?

Solution. $\ker\phi$ is evidently a vector space, and $\forall X \in \ker\phi, Y \in \mathfrak{g}$, we have $\phi([X,Y]) = [\phi(X), \phi(Y)] = 0 \Rightarrow [X,Y] \in \ker\phi$, thus $\ker\phi$ is an ideal. The converse is not true, since we can easily find an injection ϕ such that it is a vector space homomorphism but not a Lie algebra homomorphism

On the other hand, if the converse is true, if $\mathfrak{h}$ is an ideal in $\mathfrak{g}$, we can define Lie algebra homomorphism the quotient map $\pi : \mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$ which have kernel $\mathfrak{h}$

□

Exercise 2.9 Prove: if G is a connected Lie group and H is a connected Lie subgroup then H is normal in G if and only if $\mathrm{Lie}(H)$ is an ideal in $\mathrm{Lie}(G)$

Solution. Suppose G is a connected matrix group, $H \leq G$ is a connected subgroup, $\mathfrak{h}, \mathfrak{g}$ are their corresponding Lie algebras, then by Lie subgroup-Lie subalgebra correspondence, we know that G and H consists of elements of the form $\{e^{X_1}e^{X_2}\cdots e^{X_n} \mid X_i \in \mathfrak{g}\}$ and $\{e^{Y_1}e^{Y_2}\cdots e^{Y_m} \mid Y_i \in \mathfrak{h}\}$, to show $H \leq G$, we only need to show $e^X e^Y e^{-X} \in H, X \in \mathfrak{g}, Y \in \mathfrak{h}$, since then $e^{X_1}\cdots e^{X_n} e^Y e^{-X_n}\cdots e^{-X_1} \in H, Y \in \mathfrak{h}$ and $e^{X_1}\cdots e^{X_n} e^{Y_1}\cdots e^{Y_m} e^{-X_n}\cdots e^{-X_1}, Y_i \in \mathfrak{h}$, notice $e^X e^Y e^{-X} = e^{e^X Y e^{-X}} = e^{Ad_{e^Y}(X)} = e^{e^{ad_Y}(X)}$, where $e^{ad_Y}(X) = X + ad_Y(X) + \frac{ad_Y^2}{2}(X) + \cdots \in \mathfrak{h}$ since $\mathfrak{h}$ is an ideal in $\mathfrak{g}$ which is certainly a closed subspace because they are finite dimensional, thus $e^{e^{ad_Y}(X)} \in H$
Conversely, if $H \leq G$, then $e^X e^{tY} e^{-X} = e^{te^X Y e^{-X}} \in H, \forall t \in \mathbb{R}, X \in \mathfrak{g}, Y \in \mathfrak{h}$, thus $e^X Y e^{-X} \in \mathfrak{h}$, then $e^{tX} Y e^{-tX} \in \mathfrak{h}, \forall t \in \mathbb{R}$, so we have $\frac{d}{dt}|_{t=0}\ e^{tX} Y e^{-tX} = [X,Y] \in \mathfrak{h}$, therefore $\mathfrak{h} \leq \mathfrak{g}$ is an ideal

□

Exercise 2.10

Solution. Let $\mathfrak{g}_0 = \mathbb{R}$ be a real Lie algebra of dimension 1 with $[a,b] = 0$ for any $a, b \in \mathbb{R}$, suppose $\mathfrak{g} = \langle\langle\rangle\rangle$ is a real Lie algebra of dimension 1, then $[v,v] = 0$ because of anti-symmetry, hence $\varphi : \mathfrak{g} \to \mathfrak{g}_0, v \mapsto 1$ is a Lie algebra isomorphism, hence Lie algebra of dimension 1 is $\mathfrak{g}_0$ up to isomorphism

Let $\mathfrak{g}_{ab} = \mathbb{R}^2$ be a real Lie algebra of dimension 2 with $[e_1, e_2] = ae_1 + be_2$, suppose $\mathfrak{g} = \langle v\rangle \oplus \langle w\rangle$ is a real Lie algebra of dimension 2, with $[v,w] = av + bw$, hence $\varphi : \mathfrak{g} \to \mathfrak{g}_{ab}, v \mapsto e_1, w \mapsto e_2$ is a Lie algebra isomorphism, suppose $\phi : \mathfrak{g}_{ab} \to \mathfrak{g}_{cd}$ is an isomorphism, $\phi(e_1) = xe_1 + ye_2, \phi(e_2) = ze_1 + we_2$, then we necessarily have $A\begin{pmatrix}a\\b\end{pmatrix} = \det A\begin{pmatrix}c\\d\end{pmatrix}$, where $\det A \neq 0$, if $c = d = 0$, then $a = b = 0$, you can just take $A = I$, if c, d are not both zero, then a, b are not both zero and without loss of generality we may assume $d \neq 0$, in that case, if $b \neq 0$, we can take $A = \begin{pmatrix}\frac{b}{d} & \frac{cd-ab}{bd}\\ 0 & \frac{d}{b}\end{pmatrix}$, if $b = 0$, then $a \neq 0$, we can take $A = \begin{pmatrix}\frac{c}{a} & -\frac{a}{d}\\ \frac{d}{a} & 0\end{pmatrix}$, Therefore, there are two classes of real Lie algebras of dimension 2, namely $\mathfrak{g}_{00}$ and $\mathfrak{g}_{01}$

□

Exercise 2.11

$$\begin{pmatrix} & I\\ -I & \end{pmatrix}\begin{pmatrix}A & B\\ C & D\end{pmatrix} = \begin{pmatrix}C & D\\ -A & -B\end{pmatrix} = \begin{pmatrix}A & B\\ C & D\end{pmatrix}\begin{pmatrix} & I\\ -I & \end{pmatrix} = \begin{pmatrix}-B & A\\ -D & C\end{pmatrix} \Rightarrow \begin{cases}D = A\\ C = -B\end{cases}$$

Let $J = \begin{pmatrix} & I\\ -I & \end{pmatrix}$, then these are the matrices deonoted as $\mathfrak{g} \leq \mathfrak{gl}(2n, \mathbb{R})$ commuting with J, consider $\varphi : \mathfrak{gl}(n, \mathbb{C}) \to \mathfrak{g}, A + iB \mapsto \begin{pmatrix}A & B\\ -B & A\end{pmatrix}$ is evidently a linear map, and $[\varphi(A_1 + iB_1), \varphi(A_2 + iB_2)] =$
$\begin{pmatrix}A_1 & B_1\\ C_1 & D_1\end{pmatrix}\begin{pmatrix}A_2 & B_2\\ C_2 & D_2\end{pmatrix} - \begin{pmatrix}A_2 & B_2\\ C_2 & D_2\end{pmatrix}\begin{pmatrix}A_1 & B_1\\ C_1 & D_1\end{pmatrix} = \begin{pmatrix}[A_1,A_2]-[B_1,B_2] & [A_1,B_2]+[B_1,A_2]\\ -[A_1,B_2]-[B_1,A_2] & [A_1,A_2]-[B_1,B_2]\end{pmatrix} =$

$\varphi([A_1,A_2]-[B_1,B_2]+i([A_1,B_2]+[B_1,A_2]))=\varphi([A_1+iB_1,A_2+iB_2])$, thus φ is an isomorphism

Exercise 2.12

(a)

Define group homomorphism $\varphi:\mathbb{H}\to GL(2,\mathbb{C})$, $1\mapsto\begin{pmatrix}1&\\&1\end{pmatrix}$, $i\mapsto\begin{pmatrix}i&\\&-i\end{pmatrix}$, $j\mapsto\begin{pmatrix}&1\\-1&\end{pmatrix}$, $k\mapsto\begin{pmatrix}&i\\i&\end{pmatrix}$, $a+bi+cj+dk\mapsto\begin{pmatrix}a+bi&c+di\\-c+di&a-bi\end{pmatrix}$, i.e. $\lambda+j\mu\mapsto\begin{pmatrix}\lambda&\bar{\mu}\\-\mu&\bar{\lambda}\end{pmatrix}$ with determinant $|\lambda|^2+|\mu|^2=|\lambda+j\mu|^2$ which is the norm, and $\begin{pmatrix}\frac{\bar{\lambda}}{|\lambda|^2+|\mu|^2}&\frac{-\bar{\mu}}{|\lambda|^2+|\mu|^2}\\\frac{\mu}{|\lambda|^2+|\mu|^2}&\frac{\lambda}{|\lambda|^2+|\mu|^2}\end{pmatrix}$ is the left and right inverse, similarly, we can define $\Phi:GL(n,\mathbb{H})\to GL(2n,\mathbb{C})$,

$\begin{pmatrix}\lambda_{11}+j\mu_{11}&\cdots&\lambda_{1n}+j\mu_{1n}\\\vdots&\ddots&\vdots\\\lambda_{n1}+j\mu_{n1}&\cdots&\lambda_{nn}+j\mu_{nn}\end{pmatrix}\mapsto\begin{pmatrix}\lambda_{11}&\overline{\mu_{11}}&\cdots&\lambda_{1n}&\overline{\mu_{1n}}\\-\mu_{11}&\overline{\lambda_{11}}&&-\mu_{1n}&\overline{\lambda_{1n}}\\\vdots&&\ddots&\vdots&\\\lambda_{n1}&\overline{\mu_{n1}}&\cdots&\lambda_{nn}&\overline{\mu_{nn}}\\-\mu_{n1}&\overline{\lambda_{n1}}&&-\mu_{nn}&\overline{\lambda_{nn}}\end{pmatrix}$ which shows that $GL(n,\mathbb{H})$ is a matrix group of complex dimension $2n^2$, it is not compact since the entries are not bounded

(b)

$$\langle\lambda\vec{v},\vec{w}\rangle=\sum_{i=1}^{n}\overline{\lambda v_i}w_i=\sum_{i=1}^{n}\overline{\lambda}\overline{v_i}w_i=\overline{\lambda}\langle\vec{v},\vec{w}\rangle$$

$$\langle\vec{v},\vec{w}\rangle=\sum_{i=1}^{n}\overline{v_i}w_i\lambda=\langle\vec{v},\vec{w}\rangle\lambda$$

$$\langle\vec{v},\vec{w}\rangle=\sum_{i=1}^{n}\overline{v_i}w_i=\sum_{i=1}^{n}\overline{\overline{w_i}v_i}=\overline{\sum_{i=1}^{n}\overline{w_i}v_i}=\overline{\langle\vec{w},\vec{v}\rangle}$$

(c)

$\begin{pmatrix}\lambda&\bar{\mu}\\-\mu&\bar{\lambda}\end{pmatrix}\begin{pmatrix}\alpha&\bar{\beta}\\-\beta&\bar{\alpha}\end{pmatrix}=\begin{pmatrix}\lambda\alpha-\bar{\mu}\beta&\lambda\bar{\beta}+\bar{\mu}\bar{\alpha}\\-\mu\alpha-\bar{\lambda}\beta&-\mu\bar{\beta}+\bar{\lambda}\bar{\alpha}\end{pmatrix}$, we can define $\begin{pmatrix}\lambda&\bar{\mu}\\-\mu&\bar{\lambda}\end{pmatrix}\cdot\begin{pmatrix}\alpha\\\beta\end{pmatrix}=\begin{pmatrix}\lambda\alpha-\bar{\mu}\beta\\\mu\alpha+\bar{\lambda}\beta\end{pmatrix}$, then the corresponding matrix would be $\begin{pmatrix}\lambda&-\bar{\mu}\\\mu&\bar{\lambda}\end{pmatrix}$, here we are identifying $\alpha+j\beta$ with $\begin{pmatrix}\alpha\\\beta\end{pmatrix}$ and $\begin{pmatrix}\alpha_1+j\beta_1\\\vdots\\\alpha_n+j\beta_n\end{pmatrix}$ with $\begin{pmatrix}\alpha_1\\\beta_1\\\vdots\\\alpha_n\\\beta_n\end{pmatrix}$, the we have $\langle\vec{v},\vec{w}\rangle=\langle\vec{v},\vec{w}\rangle_1+j\langle\vec{v},\vec{w}\rangle_2$ where $\vec{v}=\begin{pmatrix}\alpha_1\\\beta_1\\\vdots\\\alpha_n\\\beta_n\end{pmatrix},\vec{w}=\begin{pmatrix}\gamma_1\\\eta_1\\\vdots\\\gamma_n\\\eta_n\end{pmatrix}$, and $\langle\vec{v},\vec{w}\rangle_1=\sum_{i=1}^{n}(\overline{\alpha_i}\gamma_i+\overline{\beta_i}\eta_i)$, $\langle\vec{v},\vec{w}\rangle_2=\sum_{i=1}^{n}(-\overline{\beta_i}\gamma_i+\overline{\alpha_i}\eta_i)$, hence $\langle g\vec{v},g\vec{w}\rangle_1+j\langle g\vec{v},g\vec{w}\rangle_2=\langle g\vec{v},\overline{g\vec{w}}\rangle=\langle\vec{v},\vec{w}\rangle=\langle\vec{v},\vec{w}\rangle_1+j\langle\vec{v},\vec{w}\rangle_2$, $\langle,\rangle_2$ is a symplectic form with respect to vectors of the form $\vec{v}=\begin{pmatrix}\alpha_1\\\vdots\\\alpha_n\\\beta_1\\\vdots\\\beta_n\end{pmatrix},\vec{w}=\begin{pmatrix}\gamma_1\\\vdots\\\gamma_n\\\eta_1\\\vdots\\\eta_n\end{pmatrix}$ compare to $\langle,\rangle_1$, thus $g\in U(2n)\cap Sp(2n,\mathbb{C})$

Exercise 2.13

(a)

Suppose $(\vec{v},t)\in Z$, then $(\vec{v},t)*(\vec{w},u)=(\vec{v}+\vec{w},t+u+\frac{1}{2}\langle\vec{v},\vec{w}\rangle)=(\vec{v}+\vec{w},t+u+\frac{1}{2}\langle\vec{w},\vec{v}\rangle)=(\vec{v},t)*(\vec{w},u),\forall(\vec{w},u)\in H(V)$, then $-\langle\vec{w},\vec{v}\rangle=\langle\vec{v},\vec{w}\rangle=\langle\vec{w},\vec{v}\rangle\Rightarrow\langle\vec{v},\vec{w}\rangle=0,\forall\vec{w}\in V$, since $\langle,\rangle$ is non-degenerate, $\vec{v}=0$, $Z\cong\mathbb{R}$

(b)

Define $\varphi:H(V)\to V,(\vec{v},t)\mapsto\vec{v}$ which is surjective, then $\ker\varphi=Z$, hence $H(V)/Z\cong V$

(c)

From (b), we can easily see there is an exact sequence $1\to Z\to H(V)\to V\to 1$, consider $\psi:V\to H(V),\vec{v}\mapsto(\vec{v},0)$, we have $\varphi\circ\psi=id_V$, thus the exact sequence splits

(d)

Define $\phi : H(V) \to H, (\vec{v}, \vec{w}, t) \mapsto \begin{pmatrix} 1 & \vec{v}^T & t+\frac{1}{2}\vec{v}^T\vec{w} \\ 0 & I_n & \vec{w} \\ 0 & 0 & 1 \end{pmatrix}$ is an isomorphism where H is Heisenberg group and the non-degenerate symplectic form $\langle , \rangle$ is first isomorphic to $\langle (\vec{v_1}, \overrightarrow{w_1}), (\vec{v_2}, \overrightarrow{w_2}) \rangle = \frac{1}{2}(\vec{v_1}^T\overrightarrow{w_2} - \overrightarrow{w_1}^T\vec{v_2})$, then

$\phi(\vec{v_1}, \overrightarrow{w_1}, t_1)\phi(\vec{v_1}, \overrightarrow{w_1}, t_1) = \begin{pmatrix} 1 & \overrightarrow{v_1}^T & t_1+\frac{1}{2}\overrightarrow{v_1}^T\overrightarrow{w_1} \\ 0 & I_n & \overrightarrow{w_1} \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & \overrightarrow{v_2}^T & t_2+\frac{1}{2}\overrightarrow{v_2}^T\overrightarrow{w_2} \\ 0 & I_n & \overrightarrow{w_2} \\ 0 & 0 & 1 \end{pmatrix} =$

$\begin{pmatrix} 1 & \overrightarrow{v_1}^T+\overrightarrow{v_2}^T & t_1+\frac{1}{2}\overrightarrow{v_1}^T\overrightarrow{w_1}+t_2+\frac{1}{2}\overrightarrow{v_2}^T\overrightarrow{w_2}+\overrightarrow{v_1}^T\overrightarrow{w_2} \\ 0 & I_n & \overrightarrow{w_1}+\overrightarrow{w_2} \\ 0 & 0 & 1 \end{pmatrix} =$

$\begin{pmatrix} 1 & \overrightarrow{v_1}^T+\overrightarrow{v_2}^T & t_1+t_2+\frac{1}{2}\langle(\overrightarrow{v_1},\overrightarrow{w_1}),(\overrightarrow{v_2},\overrightarrow{w_2})\rangle+\frac{1}{2}(\overrightarrow{v_1}+\overrightarrow{v_2})^T(\overrightarrow{w_1}+\overrightarrow{w_2}) \\ 0 & I_n & \overrightarrow{w_1}+\overrightarrow{w_2} \\ 0 & 0 & 1 \end{pmatrix} =$

$\phi(\vec{v_1}+\vec{v_2}, \overrightarrow{w_1}+\overrightarrow{w_2}, t_1+t_2+\frac{1}{2}\langle(\vec{v_1}, \overrightarrow{w_1}), (\vec{v_2}, \overrightarrow{w_2})\rangle)$

Exercise 2.14

(a)

Let E_{ij} denote the matrix with only the (i,j)-th entry 1, and 0 else where, then $E_{ij}E_{kl} = \delta_{jk}E_{il}$, $[E_{ij}, E_{kl}] = E_{ij}E_{kl} - E_{kl}E_{ij} = \delta_{jk}E_{il} - \delta_{li}E_{kj}$, and $\mathfrak{n}$ is spaned by $\left\{E_{ij}\right\}_{j>i}$, it is easy to see that $[\mathfrak{n}, \mathfrak{n}]$ is spaned by $\left\{E_{ij}\right\}_{j>i+1}$

(b)

Consider $\left\{I + aE_{ij}\right\}_{j>i}$ Notice $(I + aE_{ij})^{-1} = I - aE_{ij}$, and it is easy to calculate that given $j > i, l > k$, we have $(I + aE_{ij})(I + bE_{kl})(I + aE_{ij})^{-1}(I + bE_{kl})^{-1} = I + ab\delta_{jk}E_{il} - ab\delta_{il}E_{kj}$, as the same analysis in (a), we know the commutator subgroup contains all the elements in $I + [\mathfrak{n}, \mathfrak{n}]$, on the other hand, for any $I - P \in N$ with P nilpotent, we have $(I-P)^{-1} = I + P + P^2 + \cdots$, thus $(I-P)(I-Q)(I-P)^{-1}(I-Q)^{-1} = (I-P)(I-Q)(I+P+P^2+\cdots)(I+Q+Q^2+\cdots) = (I-P-Q+PQ)(I+P+Q+P^2+Q^2+PQ+\cdots) = I + [P,Q] + \cdots$, with the result in (a), we know that $[P,Q] + \cdots \in [\mathfrak{n}, \mathfrak{n}]$, and for $X, Y \in [\mathfrak{n}, \mathfrak{n}]$, $(I+X)(I+Y) = I + (X + Y + XY) \in I + [\mathfrak{n}, \mathfrak{n}]$, thus the commutator subgroup $\{N, N\}$ of N is $I + [\mathfrak{n}, \mathfrak{n}]$, $\log(e^X e^Y) = X + Y + S$, where $S \in [\mathfrak{n}, \mathfrak{n}]$, thus $f(e^X e^Y) = e^{\phi(X+Y+S)} = e^{\phi(X)+\phi(Y)} = f(e^X)f(e^Y)$ meaning f is a homomorphism

(c)

The sufficient and necessary condition needed is $\{N, N\} \leq \ker f$, first notice this map is well defined, that N is the image of $\mathfrak{n}$ under exponential map because you can take log where the power series has only finitely many terms, and $\mathfrak{n}$ is the Lie algebra of N, if $e^X = e^Y$ where $X, Y \in \mathfrak{n}$, then $X - Y \in [\mathfrak{n}, \mathfrak{n}] \leq \ker \phi$, thus $f(e^X) = e^{\phi(X)} = e^{\phi(Y)} = f(e^Y)$, the condition is obvious necessary, on the other hand, if $\{N, N\} \leq \ker f$, $e^X e^Y = I + X + Y + R$ where $R \in [\mathfrak{n}, \mathfrak{n}]$

Exercise 2.15

For any $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{R})$, suppose $g = kan$ with $k \in SO(2, \mathbb{R}), a = \begin{pmatrix} e^x & \\ & e^{-x} \end{pmatrix}, n = \begin{pmatrix} 1 & y \\ & 1 \end{pmatrix}$, then we would have $\begin{pmatrix} a^2+c^2 & ab+cd \\ ab+cd & b^2+d^2 \end{pmatrix} = g^T g = n^T a^T k^T kan = n^T a^T an = \begin{pmatrix} e^{2x} & ye^{2x} \\ ye^{2x} & y^2e^{2x}+e^{-2x} \end{pmatrix}$ thus we would necessarily have $x = \frac{\ln(a^2+c^2)}{2}, y = \frac{ab+cd}{a^2+c^2}$, and $k = g^{-T}n^T a^T$ this shows the uniqueness, only need to check that k is indeed in $SO(2, \mathbb{R})$ and calculation shows that $k^T k = ang^{-1}g^{-T}n^T a^T = \begin{pmatrix} (ad-bc)^2 & 0 \\ 0 & 1 \end{pmatrix} = I$

Exercise 2.16

(a)

Since $\mathrm{Cent}_G(hgh^{-1}) = h\mathrm{Cent}_G(g)h^{-1}$, up to conjugacy, if $g = rI_3$, then $\mathrm{Cent}_G(g) = SL(3,\mathbb{C})$, if $g = \mathrm{diag}(rI_2, a)$ where $a \neq r$, then $\mathrm{Cent}_G(g) = \{\mathrm{diag}(A, b)\}$, and if $g = \mathrm{diag}(a, b, c)$, where a, b, c are distinct, then $\mathrm{Cent}_G(g) = \mathrm{diag}(\alpha, \beta, \gamma)$, only need to prove $SL(n,\mathbb{C})$ is connected. Notice any element of $SL(n,\mathbb{C})$ can be written as $g = C\begin{pmatrix}\lambda_1 & & * \\ & \ddots & \\ & & \lambda_n\end{pmatrix}C^{-1}$, where $\lambda_1 \cdots \lambda_n = 1$, let $\begin{pmatrix}\lambda_1 & & * \\ & \ddots & \\ & & \lambda_n\end{pmatrix} = \mathrm{diag}(\lambda_1, \cdots, \lambda_n) + N$, N is nilpotent, consider path $g(t) = C(\mathrm{diag}(\lambda_1(t), \cdots, \lambda_{n-1}(t), \lambda_n(t)) + tN)C^{-1}$, where $\lambda_j(t) = r_j^t e^{it\theta_j}$, with $\lambda_j = r_j e^{i\theta_j}, 1 \leq j \leq n-1$, and $\lambda_n(t) = \frac{1}{\lambda_1(t)\cdots\lambda_{n-1}(t)}$, then $g(0) = I, g(1) = g$

(b)

Z only consists of three elements $I_3, \omega I_3, \omega^2 I_3$, where $\omega = e^{\frac{2\pi i}{3}}$, up to conjugacy, if $g = rI_3$, then $\mathrm{Cent}_G(g) = PSL(3,\mathbb{C})$ which has trivial component group, if $g = \mathrm{diag}(rI_2, a)$ where $a \neq r$, then $\mathrm{Cent}_G(g) = \{\mathrm{diag}(A, b)\}$ which has trivial component group, if $g = \mathrm{diag}(a, b, c)$, where a, b, c are distinct, suppose $a = b\omega = c\omega^2$(or $a = c\omega = b\omega^2$), then $\mathrm{Cent}_G(g)$ consists of diagonal matrices $\begin{pmatrix}x & & \\ & y & \\ & & z\end{pmatrix}$ and matrices of the form $\begin{pmatrix} & x & \\ & & y \\ z & & \end{pmatrix}$ and $\begin{pmatrix} & & x \\ y & & \\ & z & \end{pmatrix}$, and they form two different connected components other than the diagonal matrices easily seen from the fact that they are the preimages of $\{1\}$ under the maps $\begin{pmatrix}a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33}\end{pmatrix} \mapsto a_{11}a_{22}a_{33}$, $\begin{pmatrix}a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33}\end{pmatrix} \mapsto a_{12}a_{23}a_{31}$ and $\begin{pmatrix}a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33}\end{pmatrix} \mapsto a_{13}a_{21}a_{32}$, these are all connected since you can find a path connected with $\begin{pmatrix}1 & & \\ & 1 & \\ & & 1\end{pmatrix}, \begin{pmatrix} & 1 & \\ & & 1 \\ 1 & & \end{pmatrix}$ and $\begin{pmatrix} & & 1 \\ 1 & & \\ & 1 & \end{pmatrix}$ correspondingly as above, so the component group is $\mathbb{Z}/3\mathbb{Z}$, for other possibilities of a, b, c, $\mathrm{Cent}_G(g) = \mathrm{diag}(\alpha, \beta, \gamma)$ which is connected

2.3 Homework3

Exercise 2.17 Give a list of all three-dimensional and four-dimensional compact, connected Lie groups, up to isomorphism

Solution. We know the following facts:
The Lie algebra a compact Lie group is reductive
If $\mathfrak{g}$ is semisimple Lie algebra with $\dim \mathfrak{g} \leq 4$, then $\mathfrak{g} = \mathfrak{sl}(2,\mathbb{C})$
Suppose $\mathfrak{g}$ is a complex, semisimple Lie algebra, then there exists unique up to isomorphism a compact simply connected Lie group G such that $\mathfrak{g} = \mathrm{Lie}(G) \otimes_{\mathbb{R}} \mathbb{C}$
If G is a 3 dimensional compact connected Lie group, let $\tilde{G}$ be its universal cover, then $\mathrm{Lie}(\tilde{G})_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$ or $\mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C}$, in the first case, $\tilde{G}$ is necessarily isomorphic to $SU(2,\mathbb{C})$ since $\mathfrak{su}(2,\mathbb{C})_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$, and since $Z(SU(2,\mathbb{C})) = \{\pm I\}$, thus G can only be $SU(2,\mathbb{C})$ or $PSU(2,\mathbb{C})$, in the second case, $\tilde{G}$ is $\mathbb{R}^3$, hence $G = \mathbb{R}^3/\Lambda \cong T^3$
If G is a 4 dimensional compact connected Lie group, let $\tilde{G}$ be its universal cover, then since $\mathrm{Lie}(\tilde{G})_{\mathbb{C}}$ is reductive, so it is necessarily $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathbb{C}$ or $\mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C}$, in the first case, $\tilde{G} \cong SU(2,\mathbb{C}) \times \mathbb{R}$ since it has $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathbb{C}$ as its complexified Lie algebra, thus G is $SU(2,\mathbb{C}) \times T^1$ or $PSU(2,\mathbb{C}) \times T^1$, in the second case, $\tilde{G}$ is $\mathbb{R}^4$, $G = T^4$

□

Exercise 2.18

(a)

Denote $X \coloneqq (X_1, \cdots, X_n)^T$, $Y \coloneqq (Y_1, \cdots, Y_n)^T$, suppose $X' = (X_1', \cdots, X_n')^T = AX$, $Y' = (Y_1', \cdots, Y_n')^T = BY$ such that Y' is also the dual basis of X', where $A = (\alpha_{ij}), B = (\beta_{ij})$, then $\delta_{ij} = (X_i', Y_j') = \sum_k \alpha_{ik}\beta_{jk}$, thus $AB^T = I$, which also implies $B^T A = I$, hence

$$\begin{aligned}
\sum_i \pi(X_i')\pi(Y_i') &= \sum_{i,k,l} \alpha_{ik}\beta_{il}\pi(X_k)\pi(X_l) \\
&= \sum_{k,l} \left(\sum_i \alpha_{ik}\beta_{il} \right) \pi(X_k)\pi(X_l) \\
&= \sum_{k,l} \delta_{kl}\pi(X_k)\pi(X_l) \\
&= \sum_k \pi(X_k)\pi(X_k)
\end{aligned}$$

Thus Ω is independent of the choice of basis

(b)

Since $(,)$ is nondegenerate, we can choose basis X_i such that $(X_i, X_i) = \varepsilon_i$, where $\varepsilon_i = \begin{cases} 1, 1 \leq i \leq p \\ -1, p+1 \leq i \leq p+q \end{cases}$, and $(X_i, X_j) = 0, i \neq j$, then $Y_i = \varepsilon_i X_i$, $\Omega = \sum_i \pi(X_i)\pi(Y_i) = \sum_i \varepsilon_i \pi(X_i)^2$, let $[X_i, X_j] = \sum_k c_k^{ij} X_k$, c_k^{ij} are the structure constants, notice $\varepsilon_k c_k^{ij} = ([X_i, X_j], X_k) = (X_i, [X_j, X_k]) = \varepsilon_i c_i^{jk}$, so we have

$$c_k^{ij} = -c_k^{ij}$$

$$\varepsilon_k c_k^{ij} = \varepsilon_i c_i^{jk} = \varepsilon_j c_j^{ki}$$

now let's compute

$$
\begin{aligned}
[\Omega, \pi(X_j)] &= \sum_i \varepsilon_i \pi(X_i)^2 \pi(X_j) - \sum_i \varepsilon_i \pi(X_j)\pi(X_i)^2 \\
&= \sum_i \varepsilon_i \pi(X_i)\big([\pi(X_i), \pi(X_j)] + \pi(X_j)\pi(X_i)\big) + \sum_i \varepsilon_i \big([\pi(X_i), \pi(X_j)] - \pi(X_i)\pi(X_j)\big)\pi(X_i) \\
&= \sum_i \varepsilon_i \pi(X_i)[\pi(X_i), \pi(X_j)] + \sum_i \varepsilon_i [\pi(X_i), \pi(X_j)]\pi(X_i) \\
&= \sum_i \varepsilon_i \pi(X_i)\pi\big([X_i, X_j]\big) + \sum_i \varepsilon_i \pi\big([X_i, X_j]\big)\pi(X_i) \\
&= \sum_i \varepsilon_i \pi(X_i)\pi\left(\sum_k c_k^{ij}\pi(X_k)\right) + \sum_i \varepsilon_i \pi\left(\sum_k c_k^{ij}\pi(X_k)\right)\pi(X_i) \\
&= \sum_{i,k} \varepsilon_i c_k^{ij} \pi(X_i)\pi(X_k) + \sum_{i,k} \varepsilon_i c_k^{ij} \pi(X_k)\pi(X_i) \\
&\overset{(2)}{=} \sum_{i,k} \varepsilon_i c_k^{ij} \pi(X_i)\pi(X_k) + \sum_{i,k} \varepsilon_i \varepsilon_j \varepsilon_k c_j^{ki} \pi(X_k)\pi(X_i) \\
&= \sum_{i,k} \varepsilon_i c_k^{ij} \pi(X_i)\pi(X_k) + \sum_{k,i} \varepsilon_i \varepsilon_j \varepsilon_k c_j^{ik} \pi(X_i)\pi(X_k) \\
&\overset{(1)}{=} \sum_{i,k} \varepsilon_i c_k^{ij} \pi(X_i)\pi(X_k) - \sum_{k,i} \varepsilon_i \varepsilon_j \varepsilon_k c_j^{ki} \pi(X_i)\pi(X_k) \\
&\overset{(2)}{=} \sum_{i,k} \varepsilon_i c_k^{ij} \pi(X_i)\pi(X_k) - \sum_{k,i} \varepsilon_i c_k^{ij} \pi(X_i)\pi(X_k) \\
&= 0
\end{aligned}
$$

Thus Ω commutes with $\pi(X)$ for any $X \in \mathfrak{g}$

(c)

For this part, we assume $\mathfrak{g} = \mathfrak{sl}(2, \mathbb{C})$, pick $X_1 = x = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, X_2 = y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, X_3 = h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ be the basis of $\mathfrak{g}$, the its Cartan matrix is $\begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 8 \end{pmatrix}$, then we have $Y_1 = \frac{y}{4}, Y_2 = \frac{x}{4}, Y_3 = \frac{h}{8}$, $\Omega = \frac{1}{4}\pi(x)\pi(y) + \frac{1}{4}\pi(y)\pi(x) + \frac{1}{8}\pi(h)^2$

Since the irreducible representation of $\mathfrak{sl}(2, \mathbb{C})$ of dimension n is $S^{n-1}(\mathbb{C}^2)$, unique up to isomorphism, let $t_1 = (1, 0)^T, t_2 = (0, 1)^T$ be a basis of $\mathbb{C}^2$, then $t_1^{n-1}, t_1^{n-2}t_2, \cdots, t_1 t_2^{n-2}, t_2^{n-1}$ is a basis of $S^{n-1}(\mathbb{C}^2)$, we have $\Omega t_1 = \frac{3}{8}t_1, \Omega t_2 = \frac{3}{8}t_2$, thus $\Omega t_1^k = \frac{3k}{8}t_1^k$ by induction $\Omega t_1^{k+1} = \Omega\big(t_1^k t_1\big) = t_1^k \Omega t_1 + t_1 \Omega t_k = \frac{3}{8}t_1^{k+1} + \frac{3k}{8}t_1^{k+1} = \frac{3(k+1)}{8}t_1^{k+1}$, hence $\Omega(t_1^k t_2^{m-k}) = t_1^k \Omega t_2^{n-1-k} + t_2^{n-1-k}\Omega t_1^k = \frac{3(n-1)}{8}t_1^k t_2^{n-1-k}$, thus Ω acts by multiplying a scalar $\frac{3(n-1)}{8}$

Exercise 2.19

Let $\Omega = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$, then $\{X \in SL(2n, \mathbb{C}) \mid X^T \Omega X = \Omega\}$ is conjugate to $SO(2n, \mathbb{C})$, thus then also induce isomorphic Lie algebra, hence we can identify $\mathfrak{so}(2n, \mathbb{C})$ with $\{X \in M(2n, \mathbb{C}) \mid \Omega X^T + X\Omega = 0\}$ which is the same as $\left\{\begin{pmatrix} A & B \\ C & -A^T \end{pmatrix} \in M(2n, \mathbb{C}) \;\middle|\; B^T = -B, C^T = -C\right\} =: \mathfrak{g}$, then one Cartan subalgebra of $\mathfrak{g}$ will be
$\mathfrak{h} = \left\{\begin{pmatrix} D & 0 \\ 0 & -D \end{pmatrix} \in M(2n, \mathbb{C}) \;\middle|\; D = \operatorname{diag}(d_1, \cdots, d_n)\right\}$, note that

$$
\left[\begin{pmatrix} D & 0 \\ 0 & -D \end{pmatrix}, \begin{pmatrix} E_{ij} & 0 \\ 0 & -E_{ji} \end{pmatrix}\right] = (d_i - d_j)\begin{pmatrix} E_{ij} & 0 \\ 0 & -E_{ji} \end{pmatrix}
$$

$$
\left[\begin{pmatrix} D & 0 \\ 0 & -D \end{pmatrix}, \begin{pmatrix} 0 & E_{ij} \\ 0 & 0 \end{pmatrix}\right] = (d_i + d_j)\begin{pmatrix} 0 & E_{ij} \\ 0 & 0 \end{pmatrix}
$$

$$\left[\begin{pmatrix} D & 0 \\ 0 & -D \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ E_{ij} & 0 \end{pmatrix}\right] = -(d_i + d_j)\begin{pmatrix} 0 & 0 \\ E_{ij} & 0 \end{pmatrix}$$

Define $e_i \in \mathfrak{h}^*$ with $e_i\left(\begin{pmatrix} D & 0 \\ 0 & -D \end{pmatrix}\right) = d_i$, then the roots are $\Delta = \{\pm(e_i - e_j), \pm(e_i + e_j) | i < j\}$, and a set of simple roots could be $S = \{e_1 - e_2, e_2 - e_3 \cdots, e_{n-1} - e_n, e_{n-1} + e_n\}$, we can check that $K\left(\begin{pmatrix} E_{ii} & 0 \\ 0 & -E_{ii} \end{pmatrix}, \begin{pmatrix} E_{jj} & 0 \\ 0 & -E_{jj} \end{pmatrix}\right) = \delta_{ij} 3(n-1)$, thus we can treat $\{e_i\}$ as an othonormal basis
Thus $\langle e_{i-1} - e_i, e_i \pm e_{i+1}\rangle\langle e_i \pm e_{i+1}, e_{i-1} - e_i\rangle = \frac{4(e_{i-1}-e_i, e_i \pm e_{i+1})^2}{(e_{i-1}-e_i, e_{i-1}-e_i)(e_i \pm e_{i+1}, e_i \pm e_{i+1})} = 1$, $(e_{i-1} - e_i, e_j \pm e_{j+1}) = 0$ for $i < j$,
and $(e_{n-1} - e_n, e_{n-1} + e_n) = 0$, hence the root system of $\mathfrak{so}(2n, \mathbb{C})$ is of type D_n

•

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By abuse of notation, let $\Omega = \begin{pmatrix} 0 & I & 0 \\ I & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, then $\left\{X \in SL(2n+1, \mathbb{C}) \;\middle|\; X^T \Omega X = \Omega\right\}$ is conjugate to $SO(2n+1, \mathbb{C})$, hence we can identify $\mathfrak{so}(2n+1, \mathbb{C})$ with $\left\{X \in M(2n+1, \mathbb{C}) \;\middle|\; \Omega X^T + X\Omega = 0\right\}$ which is the same as
$\left\{\begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & -A_{11}^T & A_{23} \\ -A_{23}^T & -A_{13}^T & 0 \end{pmatrix} \in M(2n, \mathbb{C}) \;\middle|\; A_{12}^T = -A_{12}, A_{21}^T = -A_{21}\right\} =: \mathfrak{g}$, then one Cartan subalgebra of $\mathfrak{g}$ will be
$\mathfrak{h} = \left\{\begin{pmatrix} D & 0 & 0 \\ 0 & -D & 0 \\ 0 & 0 & 0 \end{pmatrix} \in M(2n+1, \mathbb{C}) \;\middle|\; D = \mathrm{diag}(d_1, \cdots, d_n)\right\}$, note that

$$\left[\begin{pmatrix} D & 0 & 0 \\ 0 & -D & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} E_{ij} & 0 & 0 \\ 0 & -E_{ji} & 0 \\ 0 & 0 & 0 \end{pmatrix}\right] = (d_i - d_j)\begin{pmatrix} E_{ij} & 0 & 0 \\ 0 & -E_{ji} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\left[\begin{pmatrix} D & 0 & 0 \\ 0 & -D & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & E_{ij} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\right] = (d_i + d_j)\begin{pmatrix} 0 & E_{ij} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\left[\begin{pmatrix} D & 0 & 0 \\ 0 & -D & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ E_{ij} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\right] = -(d_i + d_j)\begin{pmatrix} 0 & 0 & 0 \\ E_{ij} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\left[\begin{pmatrix} D & 0 & 0 \\ 0 & -D & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & e_i \\ 0 & 0 & 0 \\ 0 & -e_i^T & 0 \end{pmatrix}\right] = d_i\begin{pmatrix} 0 & 0 & e_i \\ 0 & 0 & 0 \\ 0 & -e_i^T & 0 \end{pmatrix}$$

$$\left[\begin{pmatrix} D & 0 & 0 \\ 0 & -D & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & e_i \\ -e_i^T & 0 & 0 \end{pmatrix}\right] = -d_i\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & e_i \\ -e_i^T & 0 & 0 \end{pmatrix}$$

Define $\delta_i \in \mathfrak{h}^*$ with $\delta_i\left(\begin{pmatrix} D & 0 \\ 0 & -D \end{pmatrix}\right) = d_i$, then the roots are $\Delta = \{\pm(\delta_i - \delta_j), \pm(\delta_i + \delta_j) | i < j\} \cup \{\pm\delta_i\}$, and a set of simple roots could be $S = \{\delta_1 - \delta_2, \delta_2 - \delta_3 \cdots, \delta_{n-1} - \delta_n, \delta_n\}$, we can check that
$K\left(\begin{pmatrix} E_{ii} & 0 & 0 \\ 0 & -E_{ii} & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} E_{jj} & 0 & 0 \\ 0 & -E_{jj} & 0 \\ 0 & 0 & 0 \end{pmatrix}\right) = \delta_{ij}(3(n-1)+2)$, thus we can treat $\{\delta_i\}$ as an othonormal basis

Thus $\langle \delta_{i-1}-\delta_i, \delta_i-\delta_{i+1}\rangle\langle \delta_i-\delta_{i+1}, \delta_{i-1}-\delta_i\rangle = \frac{4(\delta_{i-1}-\delta_i,\delta_i-\delta_{i+1})^2}{(\delta_{i-1}-\delta_i,\delta_{i-1}-\delta_i)(\delta_i-\delta_{i+1},\delta_i-\delta_{i+1})} = 1$,
$\langle \delta_{n-1}-\delta_n, \delta_n\rangle\langle \delta_n, \delta_{n-1}-\delta_n\rangle = \frac{4(\delta_{n-1}-\delta_n,\delta_n)^2}{(\delta_{n-1}-\delta_n,\delta_{n-1}-\delta_n)(\delta_n,\delta_n)} = 2$, $(\delta_{i-1}-\delta_i, \delta_j-\delta_{j+1}) = 0$ for $i<j$, and $(\delta_{i-1}-\delta_i, \delta_n) = 0$ for $i<n$, hence the root system of $\mathfrak{so}(2n+1,\mathbb{C})$ is of type B_n

Exercise 2.20

Since the Weyl group acts on Weyl chambers freely, for any regular $\gamma \in V$, $w\gamma \neq \gamma, \forall w \in W$, suppose $w = s_{\gamma_0}$ for some regular γ_0, then H_{γ_0} must contain some regular element, which is a contradiction
Therefore, $w = s_\alpha$ for some $\alpha \in \Delta$

Exercise 2.21

Note that $\alpha^\vee \in V^*$ is defined such that $\langle \beta, \alpha^\vee\rangle = \frac{2(\beta,\alpha)}{(\alpha,\alpha)}$, let $S = \{\alpha_1, \cdots, \alpha_n\} \subseteq \Delta$ be a set of simple roots which is also a basis of V, thus the gram matrix $((\alpha_i, \alpha_j))$ will be invertible, for any $f \in V^*$, $i = 1, \cdots, n$, $\langle \alpha_i, f\rangle = \sum_j c_j \frac{2(\alpha_i,\alpha_j)}{(\alpha_j,\alpha_j)} = \sum_j \frac{2c_j}{(\alpha_j,\alpha_j)}(\alpha_i, \alpha_j)$, so for any choice of $\langle \alpha_i, f\rangle$ has a unique solution $\frac{2c_j}{(\alpha_j,\alpha_j)}$, which gives a unique solution c_j, thus $\langle \alpha_i, f\rangle = \sum_j c_j \frac{2(\alpha_i,\alpha_j)}{(\alpha_j,\alpha_j)} = \langle \alpha_i, \sum_j c_j \alpha_j^\vee\rangle$, which implies that $f = \sum_j c_j \alpha_j^\vee$, hence $V^* \leq \langle S^\vee\rangle \leq \langle \Delta^\vee\rangle \leq V^* \Rightarrow V^* = \langle \Delta^\vee\rangle$, for any $\alpha \in \Delta$, $i = 1, \cdots, n$, $\langle \alpha_i, (c\alpha)^\vee\rangle = \frac{2(\alpha_i,c\alpha)}{(c\alpha,c\alpha)} = \frac{1}{c}\frac{2(\alpha_i,\alpha)}{(\alpha,\alpha)} = \frac{1}{c}\langle \alpha_i, \alpha^\vee\rangle = \langle \alpha_i, \frac{\alpha^\vee}{c}\rangle$, thus $(c\alpha)^\vee = \frac{\alpha^\vee}{c}$, but the only multiple of α that are in Δ are precisely $\alpha, -\alpha$, hence the only multiple of $\alpha^\vee$ that are in $\Delta^\vee$ are precisely $\alpha^\vee, -\alpha^\vee$
By the definition of the inner prouct on the dual Euclidean space V^*, we know $(\alpha^\vee, \beta^\vee) = \frac{4(\alpha,\beta)}{(\alpha,\alpha)(\beta,\beta)}$, thus $\frac{2(\alpha^\vee,\beta^\vee)}{(\alpha^\vee,\alpha^\vee)} = \frac{2(\alpha,\beta)}{(\beta,\beta)} \in \mathbb{Z}$ and $s_{\alpha^\vee}(\beta^\vee) = \beta^\vee - \frac{2(\alpha^\vee,\beta^\vee)}{(\alpha^\vee,\alpha^\vee)}\alpha^\vee = \beta^\vee - \frac{2(\alpha,\beta)}{(\beta,\beta)}\alpha^\vee$, then $s_{\alpha^\vee}(\alpha^\vee) = -\alpha^\vee$ and since $(s_\alpha(\beta), s_\alpha(\beta)) = (\beta, \beta)$, we have

$$\begin{aligned}
\langle \alpha_i, s_{\alpha^\vee}(\beta^\vee)\rangle &= \langle \alpha_i, \beta^\vee\rangle - \frac{2(\alpha,\beta)}{(\beta,\beta)}\langle \alpha_i, \alpha^\vee\rangle \\
&= \frac{2(\alpha_i,\beta)}{(\beta,\beta)} - \frac{2(\alpha,\beta)}{(\beta,\beta)}\frac{2(\alpha_i,\alpha)}{(\alpha,\alpha)} \\
&= \frac{2(\alpha_i, s_\alpha(\beta))}{(\beta,\beta)} \\
&= \frac{2(\alpha_i, s_\alpha(\beta))}{(s_\alpha(\beta), s_\alpha(\beta))} \\
&= \langle \alpha_i, s_\alpha(\beta)^\vee\rangle
\end{aligned}$$

Thus $s_{\alpha^\vee}(\beta^\vee) = s_\alpha(\beta)^\vee \in \Delta^\vee$

Therefore, the dual $(V^*, \Delta^\vee)$ is also a root system

From the calculation above, we also have $\left\langle \beta^\vee, (\alpha^\vee)^\vee\right\rangle = \frac{(\alpha,\beta)}{(\beta,\beta)} = \langle \alpha, \beta^\vee\rangle$

Notice that in the construction of a Dynkin diagram, the number of edges between α and β equals $\langle \beta, \alpha^\vee\rangle\langle \alpha, \beta^\vee\rangle$, then the number of edges between $\alpha^\vee$ and $\beta^\vee$ equals $\left\langle \beta^\vee, (\alpha^\vee)^\vee\right\rangle\left\langle \alpha^\vee, (\beta^\vee)^\vee\right\rangle = \langle \alpha, \beta^\vee\rangle\langle \beta, \alpha^\vee\rangle$, but with the arrow reversed, therefore A_n, D_n are the classical root systems that are self-dual, and B_n, C_n are dual to each other

Exercise 2.22

Since

$$(\vec{v}, \vec{w}) = \sum_{\alpha\in\Delta}\langle \vec{v}, \alpha^\vee\rangle\langle \vec{w}, \alpha^\vee\rangle = \sum_{\alpha\in\Delta}\langle \vec{w}, \alpha^\vee\rangle\langle \vec{v}, \alpha^\vee\rangle = (\vec{w}, \vec{v})$$

$$(\vec{v}, \vec{v}) = \sum_{\alpha\in\Delta}\langle \vec{v}, \alpha^\vee\rangle^2 \geq 0$$

And $(\vec{v},\vec{v})=0$ iff $\langle\vec{v},\alpha^\vee\rangle=\frac{2(\vec{v},\alpha)}{(\alpha,\alpha)}=0,\forall\alpha\in\Delta$ iff $\vec{v}=0$, therefore $(,)$ is symmetric and positive definite, finally, we have

$$\begin{aligned}(s_\beta\vec{v},s_\beta\vec{w})&=\sum_{\alpha\in\Delta}\langle s_\beta\vec{v},\alpha^\vee\rangle\langle s_\beta\vec{w},\alpha^\vee\rangle\\&=\sum_{\alpha\in\Delta}\frac{4(s_\beta\vec{v},\alpha)(s_\beta\vec{w},\alpha)}{(\alpha,\alpha)}\\&=\sum_{\alpha\in\Delta}\frac{4(\vec{v},s_\beta\alpha)(\vec{w},s_\beta\alpha)}{(\alpha,\alpha)}\\&=\sum_{\alpha\in\Delta}\frac{4(\vec{v},s_\beta\alpha)(\vec{w},s_\beta\alpha)}{(s_\beta\alpha,s_\beta\alpha)}\\&=\sum_{\alpha\in\Delta}\frac{4(\vec{v},\alpha)(\vec{w},\alpha)}{(\alpha,\alpha)}\\&=(\vec{v},\vec{w})\end{aligned}$$

Thus $(,)$ is also W invariant

Exercise 2.23

Suppose (V,Δ) is a root system, $\tau:V\to V,x\mapsto -x$ is a linear map, since $\alpha\in\Delta\Rightarrow-\alpha\in\Delta$, thus $\tau(\Delta)\subseteq\Delta$, also for any $\alpha,\beta\in\Delta$, $\langle\tau\alpha,(\tau\beta)^\vee\rangle=\frac{2(\tau\alpha,\tau\beta)}{(\tau\alpha,\tau\alpha)}=\frac{2(-\alpha,-\beta)}{(-\alpha,-\alpha)}=\frac{2(\alpha,\beta)}{(\alpha,\alpha)}=\langle\alpha,\beta^\vee\rangle$, and $\tau^2=\mathrm{id}$, thus τ is an automorphism (V,Δ)

Exercise 2.24

(a)

Let $e_1=(1,0),e_2=(0,1)$ be the standard basis of $\mathbb{R}^2$, suppose the coordinates of $\vec{v_1},\vec{v_2}$ are $(r_1\cos\alpha,r_1,\sin\alpha)$ and $(r_2\cos(\theta+\alpha),r_2\sin(\theta+\alpha))$, since $s_i(\vec{v})=v-2\frac{\vec{v}\cdot\vec{v_i}}{\vec{v_i}\cdot\vec{v_i}}\vec{v_i}$, we have the matrix of s_1 and s_2 with respect to e_1,e_2 are $\begin{pmatrix}-\cos 2\alpha & -\sin 2\alpha\\ -\sin 2\alpha & \cos 2\alpha\end{pmatrix}$ and $\begin{pmatrix}-\cos 2(\theta+\alpha) & -\sin 2(\theta+\alpha)\\ -\sin 2(\theta+\alpha) & \cos 2(\theta+\alpha)\end{pmatrix}$, thus the matrix for s_1s_2 would be $\begin{pmatrix}\cos 2\theta & \sin 2\theta\\ -\sin 2\theta & \cos 2\theta\end{pmatrix}$ which means that s_1s_2 is a rotation by 2θ

(b)

In order for the group $\langle s_1,s_2\rangle$ to be finite, necessarily we need θ to be a rational multiple of π, but for any $\theta=\frac{p\pi}{q}$ where $p,q>0$ are relatively prime integers, $2\theta=\frac{p}{q}\cdot 2\pi$, if we denote $r=s_1s_2,s=s_1$, then we the relations then we have the relation $s^2=1$, $r^q=1$, and $srsr=1$, therefore $\langle s_1,s_2\rangle=\langle s,r\rangle=F(s,r)/\langle s^2,r^q,srsr\rangle\cong D_q$, where D_q is the dihedral group, the symmetry group of a regular q-gon

3 An introduction to AdS geometry

Definition 3.1 Anti de Sitter space is Lorentzian space form of constant curvature -1

Motivation.

1. Gravity: study of Lorentzian metrics which solve Einstein equation ($\Lambda = -1$) $\Rightarrow$ AdS metrics are solutions. In dimension 3, AdS metrics are the only solutions. Hence 3D gravity = study of AdS metrics(Witten, Moncrief, Andersson, Mess, Benedetti)
2. Proper isometric actions: Margulis: free groups acting properly on Minkowski space
3. On a homogeneous space G/H often you can define a matural pseudo-Riemannian structure. $G = \mathrm{SO}(n,2)$, $\mathrm{SO}(n,2)/S(O(n)\times O(2))$, $\mathrm{SO}(n,2)/SO(n-1,2)$
4. 3D gravity is Teichmuller(Mess)

Definition 3.2 $(\mathbb{R}^{n+2}, \langle,\rangle)$, $\langle x,y\rangle = x_1y_1 + \cdots + x_ny_n - x_{n+1}y_{n+1} - x_{n+2}y_{n+2}$. $\mathbb{H}^{n,1} = \{x \in \mathbb{R}^{n,2} | \langle x,x\rangle = -1\}$, $T_x(\mathbb{H}^{n,1}) = x + x^\perp$ by taking derivative, $\mathrm{Isom}_0(\mathbb{H}^{n,1}) = \mathrm{SO}(n,2)$. If $x \in \mathbb{H}^{n,1}$, then $x_{n+1}^2 + x_{n+2}^2 = 1 + x_1^2 + \cdots + x_n^2 \geq 1$, can write $(x_{n+1}, x_{n+2}) = \rho(\cos\theta, \sin\theta)$, we have $\mathbb{H}^n \times S^1 \cong \mathbb{H}^{n,1}$, $\left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \\ \rho \end{bmatrix}, e^{i\theta}\right) \leftrightarrow \begin{bmatrix} x_1 \\ \vdots \\ x_n \\ \rho\cos\theta \\ \rho\sin\theta \end{bmatrix}$

Fact 3.3 Any isometry between two connected open sets of $\mathbb{H}^{n,1}$ extends to a global isometry. Any AdS manifold is modelled on $(\mathrm{Isom}(\mathbb{H}^{n,1}), \mathbb{H}^{n,1})$

Klein model of AdS space: $\mathrm{AdS}^{n,1} = \mathbb{H}^{n,1}/\{x \sim -x\} \subseteq \mathbb{RP}^{n+1}$, $\mathrm{Isom}_0 = \mathrm{PSO}(n,2)$. $\partial\mathrm{AdS}^{n,1} = \{\langle x,x\rangle = 0\}$. Geodesics are projective lines: time like geodesics: $\langle\dot\gamma,\dot\gamma\rangle < 0$, light like, space like

Poincare model: $\mathbb{H}^n \times S^1 \to \mathbb{H}^{n,1}$, $g_{\mathrm{AdS}} = g_{\mathbb{H}^n} - \rho^2 d\theta^2$

Dimension 2+1: $V = M_2(\mathbb{R})$, $q = -\det$ is a quadratic form, $\mathbb{H}^{2,1} = \mathrm{SL}_2(\mathbb{R})$, $\mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{R})$ acts on V by $(A,B)\cdot X = AXB^{-1}$, given a group homomorphism $\mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{R}) \to \mathrm{SO}(2,2)$, $\mathrm{Isom}(\mathbb{H}^{2,1}) = \mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{R})/\pm I$, $\mathrm{AdS}^{2,1} \cong \mathrm{PSL}_2(\mathbb{R})$,

4 Large scale geometry of nilpotent and solvable Lie groups

Large scale is often captured by the notion of a quasi-isometry

Definition 4.1 A (K, C) quasi isometry(QI) is $f : (X, d_X) \to (Y, d_Y)$ such that

1. $-C + \frac{1}{K} d_X(x_1, x_2) \leq d_Y(f(x_1), f(x_2)) \leq K d_X(x_1, x_2) + C$
2. $\forall y \in Y, \exists x \in X$ such that $d_Y(y, f(x)) \leq C$

Exercise 4.2

1. Any map that is bounded distance (the difference with sup norm being finite) from a quasi-isometry is a quasi-isometry
2. The composition of quasi-isometries is a quasi-isometry
3. Every quasi-isometry has a coarse inverse (i.e if $f : X \to Y$ is a quasi-isometry then there is $\bar{f} : Y \to X$ such that $f \circ \bar{f}$ and $\bar{f} \circ f$ are bounded distance from the identity map)
4. Being quasi-isometric is an equivalence relation

Solution. For any y, pick some $x \in X$ to be $\bar{f}(y)$ such that $d_Y(y, f\bar{f}(y)) = d_Y(y, f(x)) \leq C$

1. $d_X(\bar{f}(y_1), \bar{f}(y_2)) \leq K d_Y(f\bar{f}(y_1), f\bar{f}(y_2)) + KC \leq K d_Y(y_1, y_2) + 3KC.$ $d_Y(y_1, y_2) \leq d_Y(f\bar{f}(y_1), f\bar{f}(y_2)) + d_Y(y_1, f\bar{f}(y_1)) + d_Y(f\bar{f}(y_2), y_2) \leq d_Y(f\bar{f}(y_1), f\bar{f}(y_2)) + 2C \leq K d_X(\bar{f}(y_1), \bar{f}(y_2)) + 3C$
2. $\forall x \in X$, $d_X(x, \bar{f}f(x)) \leq K d_Y(f(x), f\bar{f}f(x)) + KC \leq 2KC$

□

Note.

1. If $C = 0$, then f is biLipschitz
2. If $C = 0, K = 1$, then f is just an isometry

Example 4.3 G is a finitely generated group, S, S' are finite generating sets, then $(G, d_S) \overset{\text{QI}}{\simeq} (G, d_{S'})$, here d_S is the word metric

Exercise 4.4 Any word metric on a finitely generated group is biLipschitz (hence quasi-isometric) to any other word metric on that group

Exercise 4.5 Any finite index subgroup of a finitely generated group is quasi-isometric to its supergroup

Solution. Say $[G : H] < \infty$, consider the left cosets $\{gH\}$, pick representatives $g_1 H, \cdots, g_n H$

□

Example 4.6 (Fundamental lemma of geometric group theory) If discrete group G acts proper discontinuously and cocompactly on (X, d_X) which is a proper geodesic metric space. Then G with any word metric is QI to X, X is called a model space for G

Question 4.7

- Which spaces/finitely generated groupos are QI to each other
- Examples of (self) QI's
- What properties are preserved by QI's

Example 4.8 $\mathbb{R}$ with euclidean metric is QI to $\mathbb{R}$ with max metric

Example 4.9 $f : \mathbb{R} \to \mathbb{R}$ is QI, then it preserves two "ends"

Exercise 4.10 Verify that $f : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $f(r, \theta) = (r, \ln(1 + \theta))$ is a quasi-isometry

Focus on finitely generated nilpotent and solvable groups that have (connected and simply connected) Lie groups as model spaces

Exercise 4.11 Any Riemannian metric on G give QI metric space

Definition 4.12 $\Gamma \leq G$ discrete is a uniform lattice if G/Γ is compact

Exercise 4.13 If Γ is a uniform lattice in G, then G is a model space for Γ

Warning. Not all Lie groups have uniform lattices!

Exercise 4.14 (Hard) $\mathbb{R} \rtimes \mathbb{R}$ where $\mathbb{R}$ acts on $\mathbb{R}$ by $x \mapsto e^t x$

- Verify $\mathbb{R} \rtimes \mathbb{R}$ is solvable
- Show that $\mathbb{R} \rtimes \mathbb{R}$ does not admit any uniform lattices
- Find a metric on $\mathbb{R} \rtimes \mathbb{R}$ that makes it isometric to $\mathbb{H}^2$

Fact 4.15

- Any finitely generated nilpotent group is (virtually, i.e. has finite index subgroup) a lattice in a nilpotent Lie group
- Any finitely generated polycyclic group is (virtually) a lattice in a solvable Lie group. For example, $BS(1,n) = \langle a, b | bab^{-1} = a^n \rangle$ is solvable but not polycyclic

Exercise 4.16 (Heisenberg group) $H(\mathbb{R}) \begin{bmatrix} 1 & x & z \\ & 1 & y \\ & & 1 \end{bmatrix}$ with $x, y, z \in \mathbb{R}$, $H(\mathbb{Z})$ is a uniform lattice and both are nilpotent

Example 4.17 (Sol) $\mathbb{R}^2 \rtimes \mathbb{R}$, $\mathbb{R}$ acts on $\mathbb{R}^2$ by $(x, y) \mapsto (e^t x, e^{-t} x)$ is a solvable group with lattice $\mathbb{Z}^2 \rtimes \mathbb{Z}$, $\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}^n \begin{bmatrix} x \\ y \end{bmatrix}$

QI basics:

Theorem 4.18 (Gromov) Any finitely generated group G that is QI to a nilpotent group is (virtually) nilpotent

Note. [Ershler]Exists finitely generated group G that are QI to solvable groups that not virtually solvable

Open question: What about polycyclic groups?

Theorem 4.19 (Eskin-Fisher-Whyte) If G is a finitely generated group QI to Sol then it is (virtually) a uniform lattice in Sol

Fact 4.20 $\exp : \mathcal{N} \to N$ is a diffeomorphism, assuming N is connected and simply connected

Example 4.21 (Heisenberg group) The Lie algebra has basis $\{e_1, e_2, e_3\}$, and $[e_1, e_2] = e_3$ is the only non-trivial bracket

Exercise 4.22 Show this is the Heisenberg group

Write $\mathcal{N}_i = [\mathcal{N}, \mathcal{N}_{i-1}]$, $\mathcal{N} = V_1 \oplus V_2 \oplus \cdots \oplus V_r$, where V_i is a complement of $\mathcal{N}_i$ in $\mathcal{N}_{i-1}$, i.e. $V_i \cong \mathcal{N}_{i-1}/\mathcal{N}_i$, let $\pi_i : \mathcal{N} \to V_i$ be projection

Theorem 4.23 (Ball-Box) There exists V_i and a left invariant Riemannian metric on $\mathcal{N}$ such that $V_i \perp V_j$ for all $i \neq j$ and the induced distance satisfies: There exists $a > 1$ such that $\forall R \geq 1$, $Box(\frac{R}{a}) \subseteq B_1(R) \subseteq Box(aR)$, where $Box(R) = \{n \in N || \pi_i(n)| \leq R_i\}$

Note. $|n|_h = \sum_i |\pi_i(n)|^{\frac{1}{i}}$, $\frac{1}{d} d(0,n) \leq |n|_h \leq a r d(0,n)$

Warning: $d_h(m,n) = |m^{-1}n|_h$ doesn't give a distance

QI of N

Affine maps $\mathbb{R}^n \rtimes \mathrm{GL}(n,\mathbb{R})$ are QI's, generalize this, affine maps $N \rtimes \mathrm{Aut}(N)$ are QI's

Theorem 4.24 (Xie) Let $S_g : N \to N$, $S_g(n) = n + g(\pi_1(n))$ for some $g : V_1 \to V_r$, then S_g is QI iff $|g(x) - g(y)| \leq K|x-y| + C$. S_g is bounded distance from an affine map iff g is bounded distance from a linear map

Proof. If $|g(x) - g(y)| \leq K|x-y| + C$, then S_g is a QI
Pretend d_h is a distance

$$\begin{aligned} d_h(S_g(n_1), S_g(n_2)) &= |(-n_2 - g(\pi_1(n_2))) \star (n_1 + g(\pi_1(n_2)))|_h \\ &\leq |(-n_2)| \\ &\leq d_h(n_1, n_2) + d_h(g(\pi_1(n_1)) + g(\pi_1(n_2))) \\ &\leq d_h(n_1, n_2) + |Kd(n_1, n_2) + C|^{\frac{1}{r}} \end{aligned}$$

□

See Xie for other examples of "exotic" QI's of Heisenberg group

QI between nilpotent Lie groups

Definition 4.25 The associated graded Lie group $\mathrm{gr}(N)$ as vector space is just N, but with bracket $[x,y]_\infty = \pi_{i+j}([x,y])$, for $x \in V_i, y \in V_j$, so that $[V_i, V_j]_\infty \subseteq V_{i+j}$

Theorem 4.26 (Pansu) If N_1, N_2 are QI nilpotent Lie groups, then $\mathrm{gr}(N_1) \cong \mathrm{gr}(N_2)$

Proof. Asymptotic cone of (N,d) is $(\mathrm{gr}(N), d_{cc})$, $(N, \frac{1}{t}d) \overset{t\to\infty}{\to} (\mathrm{gr}(N), d_{cc})$. And a QI indeces a biLipschitz map of asymptotic cones. Note that even if $N = \mathrm{gr}(N)$, d_{cc} is not d, but rather Carnot-Cartheodory metric(a sub-Riemannian metric)

$$d_{cc}(p,q) = \inf_{\text{join } p \text{ to } q \text{ , the tangent lines in translates of } V_1} \mathrm{Length}(\gamma)$$

□

Shalom,Sauer: cohomological invariants can be used to distinguish certain nilpotent Lie groups up to QI with the same associated gradded Lie groups

Isenrihc-Pallier-Tessera: Dehn functions of certain lattices can distinguish some nilpotent Lie groups up to QI

Open question: Which nilpotent Lie groups are QI. Smallest open case is actually 5

QI and geometry of solvable Lie gorups

$$1 \to N \to S \to N' \to 1$$

Let $N' = \mathbb{R}$, i.e. $S \cong N \rtimes \mathbb{R}$

Example 4.27 $\mathbb{R} \rtimes \mathbb{R}$, $ds^2 = e^{-2t}dx^2 + dt^2$ by letting $y = e^t$, this is the log model of $\mathbb{H}^2$. This is negatively curved

Sol= $\mathbb{R}^2 \rtimes \mathbb{R}$, $d^2 = e^{-2t}dx^2 + e^{2t}dy^2 + dt^2$, note that $(x, y_0, t) \overset{\text{Isom}}{\cong} \mathbb{H}^2$, $(x_0, y, t) \overset{\text{Isom}}{\cong} \mathbb{H}^2$. This is foliated by negatively curved Lie groups

Question 4.28 Which $N \times \mathbb{R}$ can be made negatively curved? Which $N \rtimes \mathbb{R}$ are foliated by negatively curved? Are there others? What do QI's look like? Which are QI's

Let $A : \mathcal{N} \to \mathcal{N}$ be a derivation, then $e^{tA} : N \to N$ is a one parameter family of automorphisms. $G_A = N \rtimes \mathbb{R}$ where action is e^{tA}

Definition 4.29 If the real part of eigenvalues of A is positive, then e^{tA} is a one parameter family of dilations(this is not a standard definition, a little more broad)

We say G_A is Heintze group if e^{tA} is one parameter family of dilations. We say G_A is a purely real if eigenvalues are real

Theorem 4.30 (Heintze) All negatively curved homogeneous spaces are Heintze groups, and all Heintze groups are negative curved homogeneous spaces

Exercise 4.31 Every Heintze group is QI to a purely real one. If rA, sB have same real Jordan form, then $G_A \overset{QI}{\simeq} G_B$

Special Heintze groups:

$G_{I_n} = \mathbb{R}^n \rtimes \mathbb{R} \overset{\text{Isom}}{\cong} \mathbb{H}^{n+1}$

$G_A = H_n \rtimes \mathbb{R} \overset{\text{Isom}}{\cong} \mathbb{CH}^n$, $A = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \\ & & & 2 \end{bmatrix}$, $H_2 = H$ is the Heisenberg group

There are N such that $N \rtimes \mathbb{R} \overset{\text{Isom}}{\cong}$ for any rank one symmetric space

Negative curvature give δ hyperbolic

Morse lemma: QI map geodesics to within a bounded distance of geodesics(use δ hyperbolicity)

Boundaries of G_A

For any $n \in N$, let $\gamma_n(s) = (n, s)$, this is a "vertical" geodesic choosing suitable metric

$$d_{G_A}(\gamma_n(s), \gamma_{n'}(s)) = \begin{cases} 0 & \text{as } s \to \infty \\ \infty & \text{as } s \to -\infty \end{cases}$$

$\partial G_A = N \cup \{\infty\} \cong S^n$

Fact 4.32 QI's of G_A induce homeomorphisms of the boundary(use Morse lemma)

Conjecture 4.33 (Pointed sphere - de Cornulier) Unless $G_A \overset{\text{Isom}}{\cong}$ rank 1 symmetric space, then any QI of G_A preserves ∞

Remark 4.34 True in many cases(Carraso, Xie)

We will assume all QI's fix ∞

Parabolic visual boundary metric: $\partial_\infty G_A = N$. $d_A(n, n') = e^{t_0}$ is not always a metric, but it is a quasi-metric, i.e. $d_A(n_1, n_2) \leq M(d_A(n_1, n_3) + d_A(n_3, n_2))$

$(\mathbb{R}, d_{I_1}) \simeq (\mathbb{R}, d_{Euc})$

$(\mathbb{R}^2, d_{\left[\begin{smallmatrix} 1 & \\ & 2 \end{smallmatrix}\right]}) \overset{biLipschitz}{\simeq} (\mathbb{R}), \max\left\{|\Delta x|, |\Delta y|^{\frac{1}{2}}\right\}$

Exercise 4.35 If two QI's (that fix ∞) are bounded distance apart, then they induce the same map on (N, d_A)

Definition 4.36 A QI of $G_A \cong N \rtimes \mathbb{R}$ is coarsely height respecting if it preserves level sets of $\mathbb{R}$ up to bounded distance, i.e. $f(N \times \{t_0\})$ is contained in a bounded neighborhood of $N \times \{t_0\}$

Conjecture 4.37 All QI's of non-special Heintze groups are height respecting

Exercise 4.38 Height respecting QI of G_A induce biLipschitz maps of (N, d_A)

Conclusion: $QI(G_A) \simeq biLip(N, d_A)$

Sol-like group:

$$G_{A_1, A_2} = (N_1 \times N_2) \rtimes \mathbb{R}$$

$\mathbb{R}$ acts on N_1, N_2 by $(n_1, n_2) \mapsto (e^{tA_1} n_1, e^{-tA_2} n_2)$, i.e. G_{A_1}, G_{A_2} are both Heintze groups

(EFW) Any QI of G_{A_1,A_2} is up to isometry bounded distance from $(n_1, n_2, t) \to (f_1(n_1), f_2(n_2), t)$, $f_i \in biLip(N_i, d_{A_i})$

Other example: $H \rtimes \mathbb{R} = G_A$, $A = \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}$, $[e^t x, e^{-t} y] = z$. Not much is known(open)

Conjecture 4.39 (Eskin-Fisher-Whyte) Let $E(S)$ be the group generated by exponentially distorted elements(exponential radical), then any QI coarsely preserves cosets of $E(S)$

Note. In our examples $E(G_A) = N$

Other work:

Peng $\mathbb{R}^n \rtimes \mathbb{R}^m$

Healy-Pengitore $N \rtimes \mathbb{R}^n$ CAT0

Tullia Dymarz understand subgroups of $QI(G_A)$, used to show G f.g., QI to G_A, then G is virtually a lattice of some $G_{A'}$(QI rigidity)

Exercise 4.40 (Hard) Any group that acts cocompactly and properly discontinously on a proper geodesic metric space X is finitely generated and quasi-isometric to that space

Exercise 4.41 Show that any two left invariant Riemannian metrics on a Lie group give metric spaces that are quasi-isometric

Exercise 4.42 A discrete subgroup of a Lie group that is cocompact (i.e G/H compact) is quasiisometric the Lie group itself (and is called a uniform lattice)

Exercise 4.43 There are coarsely dense subsets of $\mathbb{Z}^2$ that are not biLipschitz equivalent to $\mathbb{Z}^2$ (Burago-Kleiner)

Exercise 4.44 Are there uniform lattices in a nilpotent Lie group that are not biLipschitz equivalent?

References

Lie Groups, Lie Algebras, and Representations: An Elementary Introduction - Brian C. Hall

Introduction to Lie Algebras and Representation Theory - James E. Humphreys

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